

Lab Assignment: Skin Lesion Boundary Detection Using Canny Edge Detection

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Subject: Intro to CV Lab Task

Problem Statement

Develop a simple computer vision system that detects the boundary of a skin lesion from a skin image using image filtering and Canny edge detection. The goal is to determine how effectively edge detection can separate the lesion from the surrounding skin.

Methodology Summary

Five representative lesion images (one per class: Melanoma, Melanocytic nevus, Benign keratosis, Dermatofibroma, Vascular lesion) were processed through the following pipeline: Original → Grayscale → Gaussian Filter → Canny Edge Detection → Lesion Boundary (morphological closing + largest-contour extraction). Three Canny threshold settings (50–100, 100–200, 150–250) were tested per image, and the best threshold was selected automatically using a boundary-closure quality score (contour solidity + edge-pixel capture ratio).

Note: insert your pipeline visualization figures (Original / Grayscale / Gaussian / Canny / Boundary grid, and the three-threshold comparison grid) from the notebook output here.

Required Results Table

Image	Best Filter	Edge Method	Area (pixels)	Perimeter (pixels)
Melanoma	Gaussian	Canny 50-100	782086.0	3574.0
Melanocytic nevus	Gaussian	Canny 50-100	6133.5	4092.2
Benign keratosis	Gaussian	Canny 50-100	49424.5	3078.6
Dermatofibroma	Gaussian	Canny 50-100	393.0	133.9
Vascular lesion	Gaussian	Canny 50-100	594.5	237.9

Observation: the automatic threshold-selection step chose Canny 50-100 (the lowest/most permissive threshold pair) for all five images. As discussed under Question 5 below, this uniform choice produced a detection failure for the Melanoma image.

Final Comparison Table

Method	Noise Handling	Edge Quality	Boundary Detection	Overall Performance
Original + Sobel	Fair	Fair	Good	Fair
Original + Canny	Good	Good	Excellent	Good
Average + Sobel	Poor	Poor	Fair	Poor
Average + Canny	Fair	Poor	Poor	Poor
Gaussian + Sobel	Fair	Poor	Good	Fair
Gaussian + Canny	Fair	Poor	Poor	Poor
Median + Sobel	Fair	Fair	Good	Fair
Median + Canny	Good	Good	Good	Good

Observation: Original + Canny and Median + Canny gave the best overall performance; Average + Canny and Gaussian + Canny performed worst overall, suggesting that for this dataset, blurring before Canny removed boundary-relevant detail rather than only removing noise (consistent with the same pattern seen with transfer-learning models in Lab 02).

Questions to Answer

1. Why is Gaussian filtering applied before Canny detection?

Canny edge detection relies on intensity gradients, and raw images (especially dermoscopic photos) contain sensor/texture noise that produces many small, spurious gradient spikes. A Gaussian filter smooths out this high-frequency noise before gradient computation, so Canny detects true lesion-boundary edges rather than noise-driven false edges.

2. How did the three Canny threshold settings affect the result?

Across all five images, the automatic scoring consistently favored the lowest threshold pair, 50-100, over 100-200 and 150-250. This confirms the general pattern that lower thresholds pick up more edges and produce more contiguous, closed boundary candidates, which the solidity/capture-ratio scoring rewards. However, this came at a cost: for the Melanoma image, the low threshold was so permissive that the detected edges merged into a boundary covering almost the entire image frame rather than just the lesion, showing that 'more closed contour' is not always the same as 'correct contour.'

3. Which threshold produced the best lesion boundary?

Canny 50-100 was selected for every image by the automated scoring. It worked well for Melanocytic nevus, Benign keratosis, Dermatofibroma, and Vascular lesion, but failed for Melanoma (see Question 5). This suggests 50-100 is a reasonable default for this dataset, but a single fixed threshold is not reliable for every image, and failure cases need a fallback check.

4. Why are edges useful for detecting skin lesions?

Skin lesions typically differ from surrounding healthy skin in color and texture, which creates a measurable intensity/gradient discontinuity at the lesion's border. Edge detection isolates exactly that discontinuity, converting a diagnostic visual cue (the lesion outline) into an explicit, measurable boundary that can be used for area/perimeter calculation, shape analysis, or as an input mask for further classification.

5. What problems did you observe in detecting the lesion boundary?

Two concrete problems appear in the numeric results. First, the Melanoma image's detected area (782,086 px) is 99.97% of the full 1024x764 image frame (782,336 px) - the algorithm detected the image border itself as the 'lesion boundary' instead of the actual lesion, a clear failure case caused by the low threshold producing a fully-closed contour around the whole frame. Second, computing circularity ($4 \cdot \pi \cdot \text{Area} / \text{Perimeter}^2$) for the remaining images shows very low values - Melanocytic nevus: 0.0046, Benign keratosis: 0.0655, Dermatofibroma: 0.2754, Vascular lesion: 0.1320 - meaning the detected perimeters are far longer than a smooth lesion outline would produce. This points to jagged, fragmented, or noisy boundaries (likely from hair, texture, or incomplete edge closing) rather than clean single-lesion outlines.

6. How could your method be improved?

(1) Replace the single global Canny threshold with an adaptive/automatic method (e.g. Otsu-based or median-based auto-thresholding) computed per image rather than reusing 50-100 everywhere. (2) Add a sanity check that rejects any contour whose area exceeds a reasonable fraction of the image (e.g. >80%) or that touches the image border, to catch failures like the Melanoma case. (3) Apply hair-removal preprocessing (e.g. black-hat morphological filtering) before edge detection to reduce the jagged, low-circularity contours seen in the other four images. (4) Smooth the final contour (e.g. polygon approximation or contour smoothing) before computing perimeter, since raw pixel-level contours overestimate perimeter length. (5) Consider a learned segmentation model (e.g. U-Net) for more robust boundaries than a purely classical pipeline can provide.